Class #2 - Characteristic Equation

1. Def: The equation $\det(A - \lambda I) = 0$ is called the *characteristic equation* and the __________det $(A - \lambda I)$ is called the *characteristic polynomial.*

2. Example: $A = \begin{bmatrix} 3 & -5 & 1 \\ 1 & 0 & 2 \\ 0 & -1 & 1 \end{bmatrix}$. Find all eigenvalues of A.

   Note that λ is an eigenvalue of A __________$\det(A - \lambda I) = 0$, i.e., λ is an eigenvalue of A __________λ is a root of the __________.

3. Def: The number of times that certain λ occurs as a root of the characteristic polynomial is called the __________*multiplicity* of λ.

4. Example: Find all eigenvalue of an $n \times n$ identity matrix and state their algebraic multiplicities.

5. Note that for a real $n \times n$ matrix A,

   (a) the characteristic polynomial has the degree __________. So, A can have __________n eigenvalues (real or complex), and A can have __________n *real* eigenvalues.

(b) A is invertible __________ every eigenvalue of A is __________.

6. Suppose A is row equivalent to B. Do they have the same set of eigenvalues?

   Example: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

   Fill in the blanks:

   (a) A and its row reduced matrix __________ the same set of eigenvalues.

   (b) But, 0 is an eigenvalue of A __________ 0 is an eigenvalue of its __________.

7. Let A be a real $n \times n$ matrix, and let $\lambda_1, \lambda_2, \cdots, \lambda_k$ where k__________n are *real* distinct eigenvalues of A. Let $\vec{x}_i$ be an eigenvector corresponding to λ_i for $i = 1, \cdots, k$. Then, the set $\{\vec{x}_1, \cdots, \vec{x}_k\}$ are __________ set in $\mathbf{R^n}$. In other words, eigenvectors corresponding to __________ eigenvectors are __________.

   Proof: